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Article

Composition based on Singing Gestures: A study of gestures representing musical emotions

Abstract

Since ancient times, the creation of music has often originated from the burst of inspiration of the composer. The production of music, especially in western cultures, tends to follow its traditional music theory. With the emergence of Pythagorean music theory, Western musicology has been clearly defined and developed in depth. The composer's creation is mainly based on music creation within the framework of the Western music theory system. However, in recent years, with the rise of the topic of music gestures, some scholars have learned that music ontology as a four-dimensional hypercube is inseparable from the performer's gesture. Therefore, some musicians also try to create flexible music from the perspective of music performers and based on performance gestures. In this article, we use a recent mathematic definition of gestures and we discuss emotion from its etymological origin as a gestural movement from inside out.

Keywords: Composition; Singing; Gesture; Music Ontology; Emotion

1. Introduction

In mainstream thought in the field of music, gesture was first listed as a subject of music performance theory. This seems natural, as our bodies in many ways serve as the physical medium connecting the musical score to the acoustics of the sound. Precisely because gestures serve as a bridge between performers and musical scores, there has been extensive amounts of studies suggesting that it is a subject rich in sensibility and emotion. However, its operating logic is often neglected. In fact, the music score, as one of the basic facts of music, is like a "living fossil" that preserves the spiritual changes of the composer when contemplating about music. This is also a frozen gesture derived from imagination, which can be traced back to ancient time, as the medieval grapheme of neumes encoded gestural cues in scores. These graphemes subsequently evolved into the current symbols, which abstracts the pneumatic thread into discrete dot symbols. As Theodor Wiesengrund Adorno said in 1946, “Correspondingly the task of the interpreter would be to consider the notes until they are transformed into original manuscripts under the insistent eye of the observer; however not as images of the author’s emotion—they are also such, but only accidentally—but as the seismographic curves, which the body has left to the music in its gestural vibrations.” [2]. Therefore, apart from the influence of personal subjective factors, the creation of gestures must also be evidenced by objective logic.

Traditional composers add their own unique gestures when performing. However, this kind of creation still involves excessive subjective thinking concerns, including the structure of music design, harmony, counterpoint, etc. So, what would the acoustics be like if the composition depended entirely on the performer's gesture? This idea has been well expanded by some free jazz musicians, and some publications focusing on this music creation model have been published [1]. However, this idea has rarely been demonstrated in freely composed works by singers.

Singers are the only musicians who barely use other instruments to express music. They add many personal styles and choices when performing, and this music creation mode is very personal and subjective. In fact, most of the time, singers don’t think carefully about every note when creating music, which is affected by mood or acquired experiences. It closely resembles the concept of a behavior, and the vast majority of behavior is the result of the interaction of innated cognitive patterns and learned reactive habits. Most of the time, it is this symbiotic relationship that helps singers make choices of musical creativity [3]. However, in music as a whole, the melody created by the singer's choice can only appear in the improvisational score as a single melodic line.

This article will extend the basic geometric model of the performer's gesture and use the four-dimensional hypercube of music ontology to perform a reverse deduction, thereby demonstrating the basic logical thinking mode of the singer's free singing creation of music. The goals of this article are: (1) to prove that there is an objective logic in the emotional gestures of performers, such as singers, when they freely create works; (2) to reversely deduce the logical pattern of gestures formed by musicians through the four-dimensional structure of music ontology; (3) to discuss the impact of singing gesture on "free composition" and its possible future prospects.

2. Geometry of Gestures

Following Ref. [5], we display the diagram of gesture geometry. To the geometric concept of gestures, one of the author[Maz] refers to two mathematical structures: a digraph (directed graph) and a topological space [4]. A digraph consists of three parts: arrows (A), vertices (V) and a map (P), which are used as connections between vertices. $P(a) = [h(a), t(a)]$ where h (a) and t (a) represent the head and tail of arrows. This is shown in **Fig.1** below.

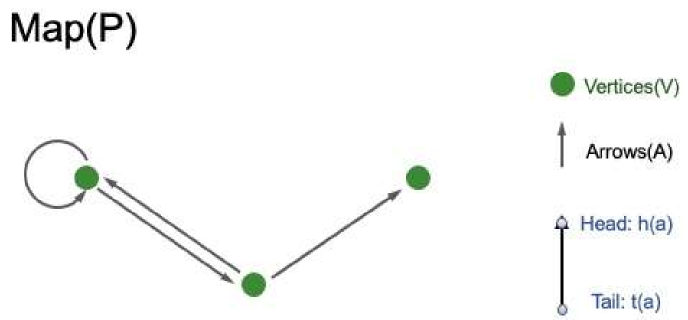

Fig. 1 | *A directed graph consisting of three vertices connected by four arrows*

A topological space is the second concept to define a gesture. For a topological space Z, in each point $z_i \in \boldsymbol{Z}$, there exists a collection of subsets $\boldsymbol{W} \subseteq \boldsymbol{Z}$, the open neighborhoods of z_i. For two neighborhoods of z_i, their intersections should also be a neighborhood of z_i. Next, we introduce an *open set*, $\boldsymbol{Q} \subseteq \boldsymbol{Z}$, which is a subset of $\boldsymbol{Z}$ that is a neighborhood for each of its element $z_i \in \boldsymbol{Q}$. The spatial diagraph $\boldsymbol{Z}^*$ of a topological space $\boldsymbol{Z}$ is the digraph, whose arrows are the continuous functions $x: [0, 1] \rightarrow \boldsymbol{Z}$ from the unit interval $[0, 1]$ of real numbers into $\boldsymbol{Z}$. The vertices of $\boldsymbol{Z}^*$ are the points of $\boldsymbol{Z}$, they are the head and tail $x(1)$ and $x(0)$ of the arrows x. A morphism between diagraph (e) can be defined when two digraphs (***R*** and $\boldsymbol{P}$) are co-present: $e\colon \boldsymbol{R} \to \boldsymbol{P}$ defines a pair of maps $e_V\colon V_{\boldsymbol{R}} \to V_{\boldsymbol{P}}$, and $e_A\colon A_{\boldsymbol{R}} \to A_{\boldsymbol{P}}$. Such an operation maps the heads and tails of the arrows to their images.

With the foregoing concepts, a *gesture* n can be defined: it is in principle a morphism $n\colon \boldsymbol{R} \to \boldsymbol{Z}^*$. $\boldsymbol{R}$ and $\boldsymbol{Z}^*$ are called the *skeleton* and *body* of a gesture, see **Fig. 2** as a sketch.

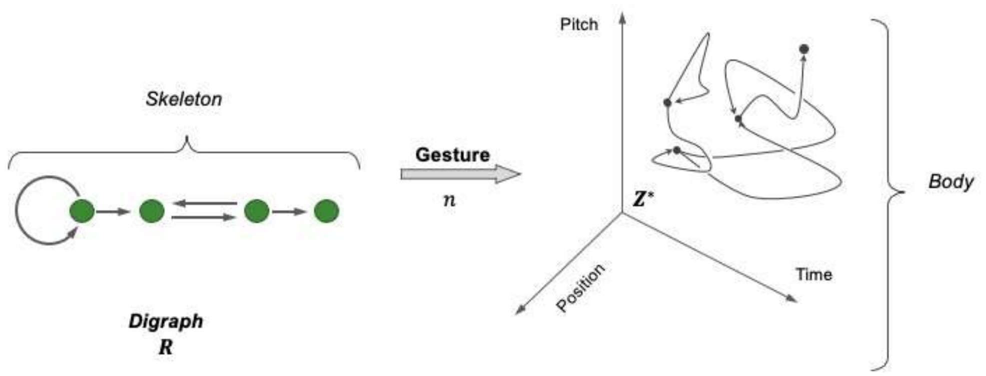

Fig. 2 | *Display of a gesture digraph in music space.*

Gesture morphisms are the key elements that relate gestures together. If $n\colon \boldsymbol{R} \to \boldsymbol{Z}^*$ and $x\colon \boldsymbol{P} \to \boldsymbol{K}^*$ are two gestures, a morphism $(y\,, e)\colon n \to x$ is a pair with $y\colon \boldsymbol{R} \to \boldsymbol{P}$ and $e^*\colon \boldsymbol{Z}^* \to \boldsymbol{K}^*$ two diagraph morphisms. e^* is derived from a continuous function $e\colon \boldsymbol{Z} \to \boldsymbol{K}$. It is also required that $n \cdot e^* = x \cdot y$. Gestures together with their morphisms are a mathematical category.

Moreover, the set $\boldsymbol{R@Z}$ of gestures from $\boldsymbol{R}$ to $\boldsymbol{Z}^*$ is again a topological space. Therefore the concept of hypergestures can also be defined: A hypergesture is a gesture $\boldsymbol{P} \to \boldsymbol{R@Z}$, in other words, are gestures of gestures.

The core result of hypergestures is the *Escher Theorem*: if $\boldsymbol{R}$ and $\boldsymbol{P}$ are both digraphs, and $\boldsymbol{Z}$ is a topological space, then the topological spaces $\boldsymbol{R@P@Z}$ and $\boldsymbol{P@R@Z}$ generated by exchanging the skeleta $\boldsymbol{R}$ and $\boldsymbol{P}$ are *isomorphic* [6].

3. The principles of gestural generation of consciousness

Gestures arise from hand and body movements that people often make when performing strenuous cognitive activities, such as speaking or problem solving. Some studies [7] have shown that such gestures are often associated with spatial information. For example, when discussing the size and shape of objects, the layout of a room, and movement in space. Therefore, many scholars believe that the creation of gesture has the function to express spatial information. McNeill's [8] and Kenton's [9] seminal theory in 1992 suggested that gestures reflect mental images. It is believed that the "growth point" behind any discourse contains image and language information, and these mental images are the basis for the creation of gestures. The theory proposed by Kita and Özyürek [10] in 2003 claims that representational gesture originates from spatio-motoric representation, which encodes non-linguistic, spatial, and motor attributed information. The theory uses information as expressed in a mechanism to produce language. The phenomenon of interaction with spatial motion is the interface to facts by gestures.

As an established fact of music, sheet music, like articles, is the result of "frozen" creative thinking. In addition to the audience obtaining the linguistic information through reading and listening, the performer usually plays a good role in interpreting the spatial information. For example, singers often consider adding their own understanding of musical facts to a new interpretation of the spatial information of these works. To achieve this effect, the singer needs to change the spatial information by changing the gesture. In other words, the singer changes the encoding gestural order of this spatial motion representation.

So, how do singers encode spatial motion representations? First, the process of encoding is often related to human subjective thinking, such as emotions. Many studies have shown that emotional states can be inferred from human facial expressions. Specific facial muscle movement configurations appear to broadcast or display a person's emotions in general, which is why they are often referred to as emotional and facial expressions [11]. Recall that the etymology of emotion, e-movere in Latin, means that in an emotion, one produces a gestural movement from inside to outside. Different expressions are like different codes corresponding to different gestures. The combination of these gestures representing different emotions makes the melody created by the singer richer in color changes.

Some scholars have also done some comparative studies and found that positive emotions tend to produce smoother physical gestures [12]. This also means that different emotions will affect the speed of transmission of gestural information. From another perspective, emotional expression not only results in the movement of the human face or body, but the very process of emotional creation is itself also a gesture. For example, when singers sing upbeat songs, they can't help but raise their arms or smile. At the same time, in the spirit of this mood, the brain will continuously repeat these gestural instructions. In this case, the role of language in conveying emotions will gradually weaken, thus strengthening the reliance on spatial information. Therefore, after becoming familiar with the codes of these positive information, the singer will sing faster and faster, or the voice will become louder and louder, to continue this emotion. This process of emotional continuation is the emotional gesture. See **Fig.3**, for three gestures starting from the same skeleton, which displays the emotional "logic". For reasons of space, we omit explanation of this logic.

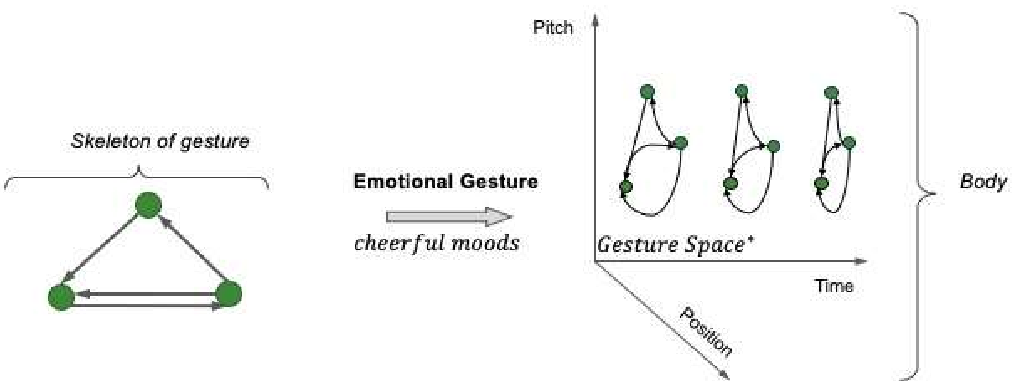

Fig.3* | *Three emotional gestures with the same skeleton (cheerful emotions)

4. The connection between Gestures and Music Ontology

According to the classical ontological landscape of music, we can interpret the ontology of music as a four-dimensional hypercube. Its layer of facts consists of three dimensions: Realities, Communication and Semiotics. Each dimension has the following three "values". First of all, the three values of Realities include (1) physical reality, that is, acoustic phenomena; (2) mental reality, that is, music theory thinking such as musical structure; (3) psychological reality, that is, emotions and feelings. Secondly, the value covered by communication was proposed by Jean Molino and Paul Valéry: (1) the creative instance(composer) of the work; (2) the nature of the created material and the actual value of the output, also called neutral level by Molino; (3) Evaluate the value based on the aesthetics of different audiences. Finally, the three levels of value included in Semiotics are: (1) expressive surface of signs; (2) deep meaning, that is, the content hidden behind; (3) the method of transitioning from the surface to the content, that is, the process of generating content [13]. The fact layer is displayed as a classical three-dimensional cube, see **Fig.4**.

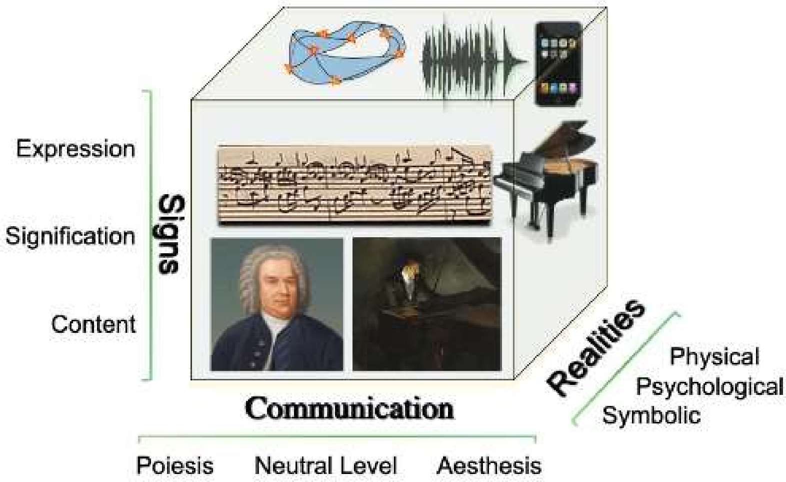

Fig.4 **|** *The classical three-dimensional cube of musical ontology.*

According to the description of Ref. [1], the hypercube of the music ontology is composed of a Gestural layer, Processes layer and Facts layer. The Processes and Facts layers are both initiated from the most basic Gestural layer, see **Fig.5**. Therefore, analyzing the composition of the

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Gesture layer may restructure the Facts layer to help composers create from a new direction: first gesture, later facts following gestures.

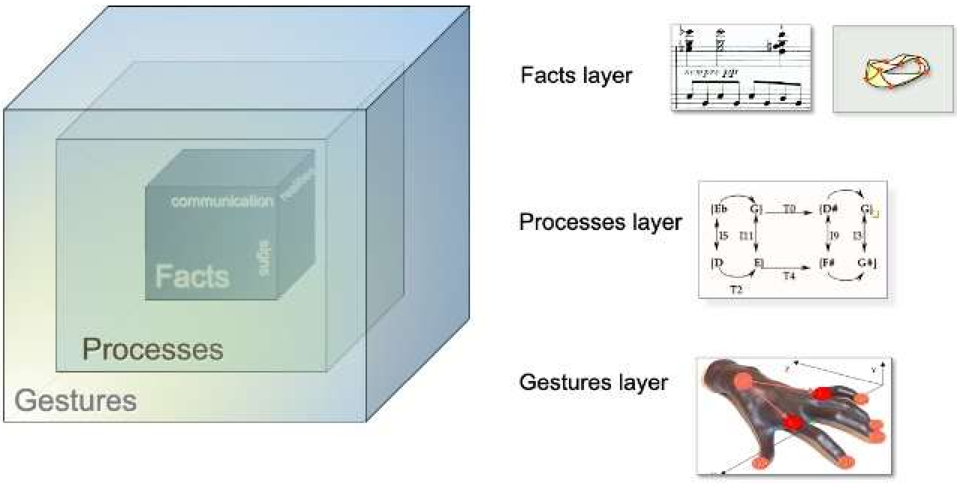

Fig.5* | *The hypercube of musical ontology.

The process layer serves as a factory that produces facts and connects them to gestures. In a popular sense, it is the process of recording how gestures generate output. See **Fig.6**.

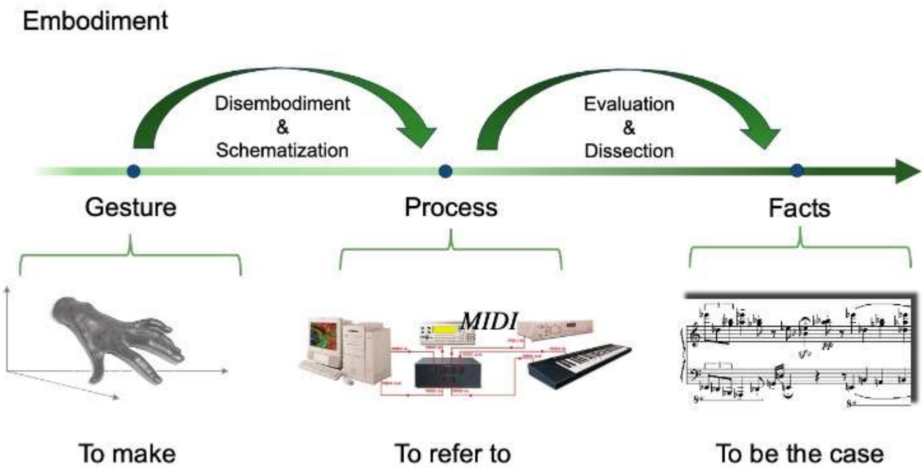

Fig.6* | *The fourth dimension of embodiment in the ontology of music.

5. Gestures applied to composition by free singing

Typically, traditional music as a fact is subject to extensive research and analysis. Composers have very mature music theory systems to help them create music. In recent years, with the popularity of the topic of music gesture, scholars have done a lot of analysis on the gestures and processes produced by musicians when performing musical facts. However, if the gestural layer forms an independent system and acts upon the factual layer, will it affect the known music theoretical system of the factual layer?

Let us discuss the critical example, the famous jazz singer Jeanne Lee has created a large number of improvised free jazz works. The creative ideas of these works hardly follow traditional Jazz theory, but are more inclined to the singer’s own feelings about the sound. As she said: “The voice is a very important instrument, it’s part of the body and can emulate bodily feelings. So using the psoas muscles and the diaphragm together you can take it into dance or voice or both you learn to work with the dynamics of the feeling you learn to work with the emotions. When your body is working you don’t have to think of a horn but you can think of body movement.” [14]. Since the 1960s, Jeanne Lee chose to stay away from traditional musical concepts, detaching music from musicality in order to increase space and quietness. She tried many ways to achieve her goal, such as reading poems with music accompaniment, paying more attention to the syllables or repeated words of the poem itself, and combining it with some special sounds to convey it to the audience. Or train herself to use dance to convey emotions to the audience through body perception. When she performed the music created through gesture, the acoustic often sounded very strange. It did not have the musical structure in traditional music theory, and the sound effect was more personal, unique and impressive.

In fact, there is a basic logic for singers to focus on the gesture of the music itself and create freely. The authors believe that such a creation can be approached from two aspects: First, it is a horizontal melody development, that is, personal improvisation based on the singer’s leading ideas. The second aspect is a secondary creation completed by the accompanying musicians in combination with the main melody, that is, a vertical musical structure. This not only requires the musicians to interpret the poems and the music sung by the singers, but also requires their own secondary thinking after interpreting the music. Therefore, regarding the second point, the musician's secondary creation can be called a hypergesture after the gesture. It is worth noting that the accompanist here can be a musician who can flexibly operate an instrument or an electronic musician using electronic equipment, such as MIDI. The gesture processing model behind this procedure should also follow a certain logic, which we discuss now.

6. The gesture layer is used for the thinking logic of composing music.

In order to understand the basic logic of the gestural layer, we first need to refer to the hypercube relationship of the music ontology. The gestural layer has an inseparable causal relationship with the most basic three-dimensional cube of music. In other words, focusing on the three dimensions of the factual layer: Realities, Communication and Semiotics, we can further infer the physical and mental changes made by the performer, following the logic of the basic thinking of the gesture. According to the previous ideas, the singer's free creation through gesture consists of two aspects, namely horizontal melody and vertical collaborative musical structure.

The creation of a single melody line is more derived from the singer's inspiration. It is a gesture that develops from consciousness and is usually spontaneous and uncontrollable. The singer's performance actually conveys some factual spatial information to the audience. In traditional music, this fact is often created by the composer first, and then the singer performs secondary processing of spatial information. However, for performers who create freely, this is based on the fact that the spatial information has already taken shape and comes from the singer's inspiration. It comes from the following three possibilities:

(1) The singer's own music education background. Usually, musicians who have experienced systematic education in traditional music theory will have more rigorous creative logic than musicians who have not received systematic music education background. Usually these musicians have many musical works in their brain memory as creative materials, so the music they create will be more or less mixed with the shadows of different types of music.

(2) The intention of the singer. This element is closely related to the different life experiences of different singers which contains two dimensions: individual intention and collective intention. Personal intention focuses more on self-reflection within the individual. Self-reflection is the individual's thinking after the fact. It includes emotional changes and the understanding of right and wrong. Music and language have very similar aspects, and the concept of communication arises when language occurs in the context of more than one individual. Communication is always a collective endeavor that requires at least two interlocutors to coordinate their individual contributions and fulfill their respective communication responsibilities. In the gesture creation and thinking of singers' performances, they serve more as a topic guide to communicate with the accompaniment musicians and even the audience. George Herbert Mead once said:"We are not… building up the behavior of the social group in terms of the behavior of the separate individuals composing it; rather, we are starting out with a given social whole of complex group activity, into which we analyze (as elements) the behavior of each of the separate individuals composing it." [15]. Therefore, when an individual receives external information from the group, the individual will process the information, resulting in changes in his/her outlook on life and values. For example, reflection on society and human nature. These experiences will allow singers to perform secondary processing of lyric information when singing, conveying different meanings to the audience.

(3) The influence of the singer's emotions. To understand it from a popular perspective, it is the change of the singer's mood when singing. According to our previous ideas, we know that the formation of a singer's gesture is inseparable from one's thoughts. When we regard the gesture in the singer's consciousness as an independent emotional expression, the movement trajectory it forms is a single expression. But in the composition process, the main melody is composed of multiple different emotions, which is a smooth process of moving the music forward. Therefore, the mood in the music piece is often not static; its changes in different gestures will also change the meaning of the lyrics and music. And this is what the accompanist musicians need to understand most carefully. The horizontal single melody line, as a reality factor, is usually the most intuitively reflected in musical facts.

7. Concrete application through gesture composition.

After understanding the basic logic of the singer's use of gesture to compose music, we can find that the accompaniment musician's gesture is a response to the singer's gesture creation. Based on this point, the occurrence of the singer's gesture serves as the fundamental reason for music creation. First, the most basic element for music to occur, is time. In response to this, the singer's breathing gesture becomes a crucial part. Through some videos [22], we can find that when singing with the symphony orchestra, some singers usually open their mouths, make a surprised expression, and take a very fast breath. This hints to the conductor that they will sing at a faster speed next. In some very slow music processing [23], it can be seen that singers' bodies are often not in a very active state, and they often show a state of enjoying music on their faces. In terms of pitch changes, if the interval span of the front and back notes is large, the singer will often make preparations in advance, which can often be shown through micro-expressions. For example, when singing a high note, the opera singer will breathe fully and, before singing the high note, raise the soft palate in preparation for singing the high note [24]. In terms of volume, for improvisational compositions, there is a process of increasing or decreasing the volume. When a singer sings a passage with a strong volume, one's body will often be in a more active state to help one's voice have better explosive power to express this emotion. Quiet pieces of music are just the opposite [25].

8. Conclusion.

This article chooses to conduct an in-depth analysis based on the singer's gesture for the following two reasons: *First*, in recent years, there have been very few documents that study the connection between singing gesture theory and composition theory. There is a blank in this aspect, which prompted the emergence of this article. *Second*, gestures produced by a singer are the only performance form without the use of other instruments. Therefore, such a performance process will be more internal and directly linked to the brain's information processing. Our approach can reduce the obstacles to the direct creation of music by gesture.

From the previous discussion, we learn that music, in its three-dimensional factual ontology, is generated by the superposition of different musical gestures. Its core musical facts include musical scores, and its dynamic performative trajectory generated by the movement of gestural combinations. Such an operating logic forms the four-dimensional hypercube ontological model of music. The major conclusions can be summarized as follows:

(1) There is an objective logic to the singer's posture when freely creating works. In the logic of traditional music system, facts are used as causes and gestures are produced as results. An opposed to this approach, this article is based on gesture as the basis of creation, and reversely analyzes its results, that is, musical facts. This results in a new logical system, the logical system of gestural creation.

(2) Based on this objective logic, this article is divided into three specific steps to detail the logic. First, we derive emotional gestures based on the geometric representation of the gestural skeleta. Secondly, we know the specific structure of music ontology and learn the causal relationships at three layers in the ontology. Finally, based on the logical relationships at three layers in music ontology, we deduced the impact of the creation of singer's gesture on music facts, so as to combine it with practice to obtain the application of concrete gestures in composition.

(3)This research will have many development directions in the future. In addition to encouraging more flexible and free improvisation in music performances, it is also likely to help artificial intelligence in music creation in the future. With the development of science and technology, there is currently a lot of AI analysis of human expressions, which can generate a large amount of data about emotions through human facial micro-expressions. If the singer's creative gesture is combined with this technology, musical works dominated by the human spirit may be produced in large numbers at a faster speed.

9. Reference.

[1] Mazzola, G., Cherlin, P. B. Flow, gesture, and spaces in free jazz: Towards a theory of collaboration. Berlin, Germany: Springer Science & Business Media, 2008.

[2] Adorno, Theodor W., and Henri Lonitz. Zu Einer Theorie Der Musikalischen Reproduktion: Aufzeichnungen, Ein Entwurf Und Zwei Schemata. 1. Aufl. Frankfurt am Main: Suhrkamp, 2001.

[3] Meyer, Leonard B. Style and Music: Theory, History, and Ideology. University of Chicago Press, 1996.

[4] Mazzola, G. The Topos of Music: Geometric Logic of Concepts, Theory and Performance. Birkhauser Verlag. Edited by Stefan Göller & Stefan Müller, 2002.

[5] Mazzola, G., Andreatta, M. Diagrams, gestures and formulae in music. Journal of Mathematics and Music, 2007.

[6] Mazzola, G. The Topos of Gestures. In: The Topos of Music III: Gestures. Computational Music Science. Springer, 2017.

[7] Martha W. Alibali. Gesture in Spatial Cognition: Expressing, Communicating, and Thinking About Spatial Information, Spatial Cognition & Computation, 2005.

[8] McNeill, D. Gesture and thought. University of Chicago Press. 2005.

[9] Kendon, A. Gesture: Visible Action as Utterance. Cambridge: Cambridge University Press; 2004.

[10] Kita, S., Özyürek, A. What does cross-linguistic variation in semantic coordination of speech and gesture reveal?: Evidence for an interface representation of spatial thinking and speaking. Journal of Memory and Language, 2003.

[11] Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements. Psychological Science in the Public Interest, 2019.

[12] Dael. N, Goudbeek. M, Scherer. KR. Perceived gesture dynamics in nonverbal expression of emotion. Perception, 2013.

[13] Mazzola, G., Geometrie der Töne. Elemente der Mathematischen Musiktheorie. Basel etc., Birkhäuser Verlag, 1990.

[14] Porter, E. Jeanne Lee's Voice. Critical Studies in Improvisation, 2006.

[15] Dascal, M. Interpretation and Understanding. John Benjamins Publishing, 2003.

[16] Norgaard, M., Emerson, S. N., Dawn, K., Fidlon, J. D. Creating Under Pressure: Effects of Divided Attention on the Improvised Output of Skilled Jazz Pianists. Music Perception: An Interdisciplinary Journal, 2016.

[17] Eagle, C. T., Harsh, J. M. Elements of Pain and Music: The Aio Connection. Music Therapy. 1988

[18] Johnson-Laird, P. N. “How Jazz Musicians Improvise.” Music Perception: An Interdisciplinary Journal, 2002.

[19] Pont, G. Philosophy and Science of Music in Ancient Greece. Nexus Network Journal, 2004.

[20] Kelly, S.D., Ngo Tran, QA. Exploring the Emotional Functions of Co-Speech Hand Gesture in Language and Communication. Top Cogn Sci, 2023.

[21] Lor-Lhommet, M., Marsella, S. Expressing Emotion Through Posture. The Oxford Handbook of Affective Computing, 2014.

[22] Bartoli, C. Aria Agitata da due venti from Griselda by Antonio Vivaldi. Video. Link: https://youtu.be/2O9OfxS0r08?feature=shared. Time: 1'06"-1'08"

[23] Didonato, J. Aria When I am laid in earth from opera Dido and Aeneas by Henry Purcell. Video. Link: https://youtu.be/3jPiYzUhGDo?feature=shared. Time: from 2'08"

[24] Mas, H. Aria O ma lyre immortelle from opera Sapho by Charles Gounod. Video. Link: https://youtu.be/ZP4BtO3H0No?feature=shared. Time: 8'22"-8'28"

[25] Gheorghiu, A. Aria Vissi d'arte from opera Tosca by Giacomo Puccini. Video. Link: https://youtu.be/VRPDulYxIAc?feature=shared. Time for loud volume: 2'36"-2'45". Time for soft volume: 1'50"-2'01"